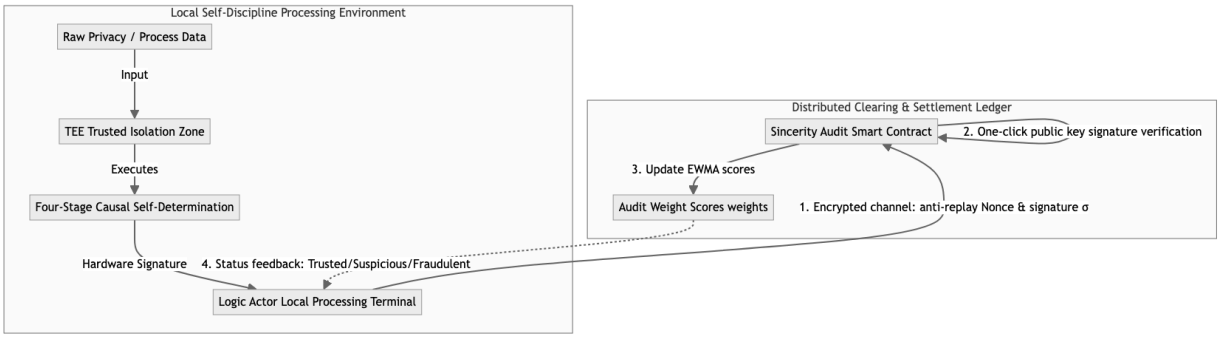

Local Self-Discipline Processing Environment
Raw Privacy / Process Data
Input
TEE Trusted Isolation Zone
Executes
Four-Stage Causal Self-Determination
Hardware Signature
4. Status feedback: Trusted/Suspicious/Fraudulent
Logic Actor Local Processing Terminal
Distributed Clearing & Settlement Ledger
Sincerity Audit Smart Contract
2. One-click public key signature verification
3. Update EWMA scores
Audit Weight Scores weights
1. Encrypted channel: anti-replay Nonce & signature σ

图1

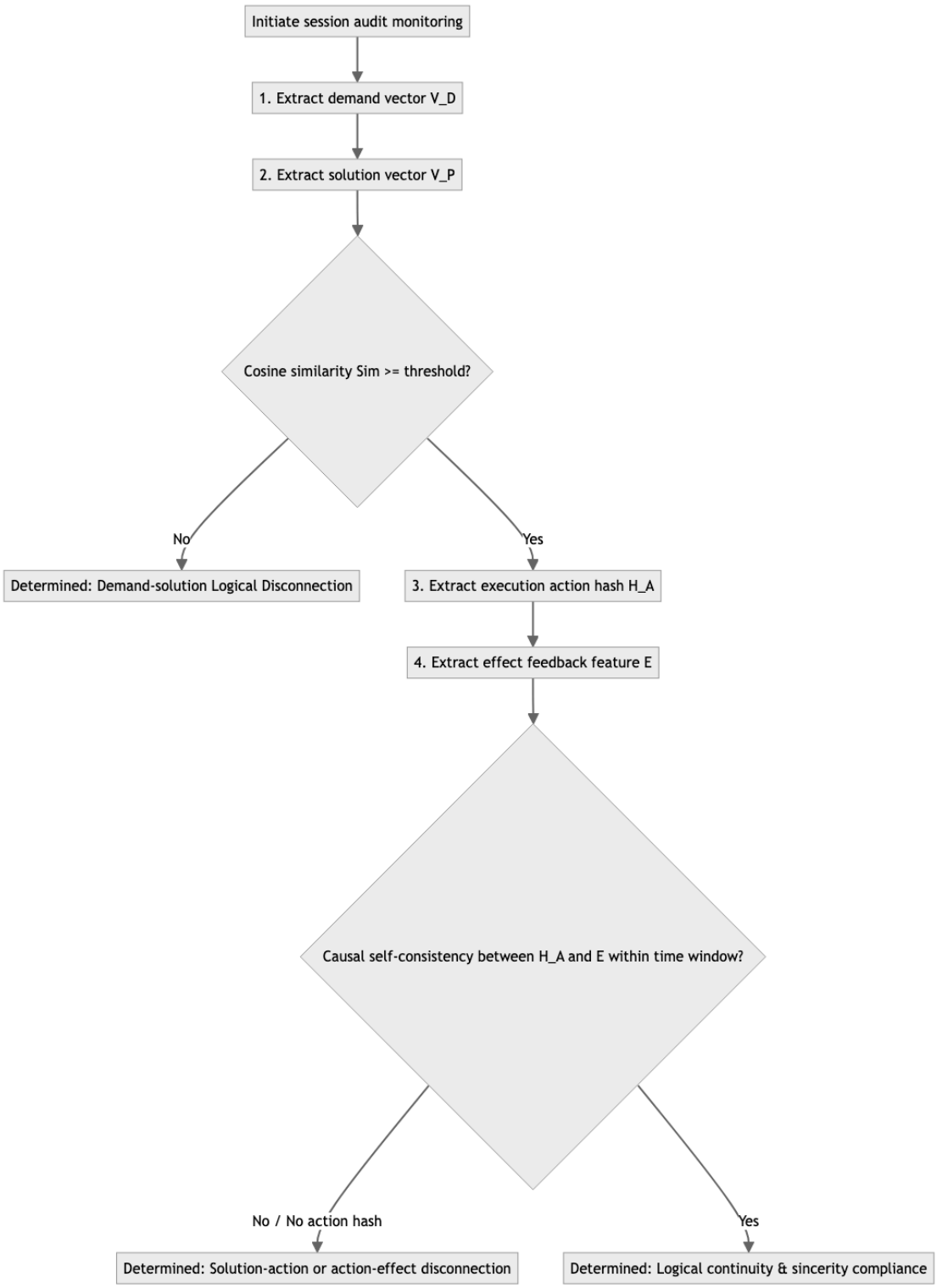

Initiate session audit monitoring
1. Extract demand vector V_D
2. Extract solution vector V_P
Cosine similarity Sim >= threshold?
No
Yes
Determined: Demand-solution Logical Disconnection
3. Extract execution action hash H_A
4. Extract effect feedback feature E
Causal self-consistency between H_A and E within time window?
No / No action hash
Yes
Determined: Solution-action or action-effect disconnection
Determined: Logical continuity & sincerity compliance

图2

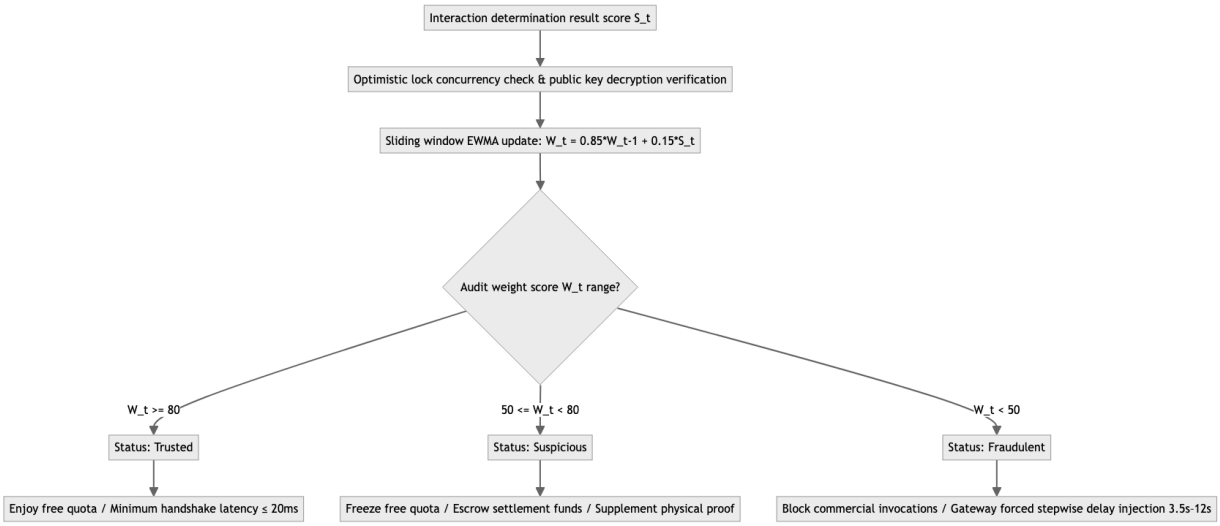

Interaction determination result score S_t
Optimistic lock concurrency check & public key decryption verification
Sliding window EWMA update: W_t = 0.85*W_t-1 + 0.15*S_t
Audit weight score W_t range?
W_t >= 80
50 <= W_t < 80
W_t < 50
Status: Trusted
Status: Suspicious
Status: Fraudulent
Enjoy free quota / Minimum handshake latency ≤ 20ms
Freeze free quota / Escrow settlement funds / Supplement physical proof
Block commercial invocations / Gateway forced stepwise delay injection 3.5s-12s

图3